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# LECTURE 7: VDW EOS showing VDW Loop
# Script: VDW-loop
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# Description:
#   This notebook plots the VDW EOS isotherms for CO2,
↪   highlighting the van der Waals loop
#   for subcritical temperatures and indicating the critical
↪   point.
#   It uses an interactive slider to adjust the temperature and
↪   visualize changes in the isotherm
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# On Google Colab, install ipympl and enable interactive widgets
import sys
if "google.colab" in sys.modules:
    %pip install -q ipympl
    from google.colab import output
    output.enable_custom_widget_manager()

import numpy as np
import matplotlib.pyplot as plt
from ipywidgets import interact, FloatSlider

def vdw_pressure(T, V_m, a, b):
    """
    Calculate Pressure using van der Waals EOS.
    T: Temperature (K)
    V_m: Molar Volume (L/mol)
    a, b: vdW constants for CO2
    """
    R = 0.08314 # L*bar/(mol*K)
    return (R * T) / (V_m - b) - a / (V_m**2)

# Constants for CO2 obtained from critical point data
a_CO2 = 3.64 # L^2 bar mol^-2
b_CO2 = 0.04267 # L mol^-1
Tc_CO2 = 304.13 # K
Pc_CO2 = 73.77 # bar

def plot_vdw_isotherm(T):
    # 1. Define Volume Range
    # We start slightly above 'b' to avoid singularity
    v = np.linspace(b_CO2 * 1.1, 1.0, 1000)

    # 2. Calculate Pressure

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P = vdW_pressure(T, v, a_CO2, b_CO2)

# 3. Plot
plt.figure(figsize=(8, 6))

# Plot the user-selected isotherm
plt.plot(v, P, linewidth=2.5, color='blue', label=f'Isotherm
↪ at T={T:.1f} K')

# Plot the Critical Isotherm (Reference)
P_crit_iso = vdW_pressure(Tc_CO2, v, a_CO2, b_CO2)
plt.plot(v, P_crit_iso, '--', color='gray', alpha=0.6,
↪ label=f'Critical Isotherm ({Tc_CO2:.1f} K)')

# Formatting
plt.ylim(0, 150) # Limit P to see the "loops" clearly
plt.xlim(0.04, 0.6)
plt.xlabel('Molar Volume $V_m$ (L/mol)', fontsize=12)
plt.ylabel('Pressure $P$ (bar)', fontsize=12)
plt.title(f'van der Waals Isotherms for CO$_2$ \n ($T_c$
↪ \approx 304 K)', fontsize=14)
plt.axhline(0, color='black', linewidth=1)

# Highlight the Critical Point
plt.plot(3*b_CO2, Pc_CO2, 'ro', label='Critical Point')

# Add descriptive text based on regime
if T > Tc_CO2:
    plt.text(0.3, 120, "Supercritical Fluid\n(Behaves like a
↪ dense gas)", fontsize=10, color='green')
elif T < Tc_CO2:
    plt.text(0.3, 120, "Subcritical\n(Phase Split
↪ Possible)", fontsize=10, color='red')
    plt.text(0.1, 20, "Look for the\n'van der Waals Loop'",
↪ fontsize=9, color='red')

plt.grid(True, linestyle='--', alpha=0.5)
plt.legend()
plt.show()

# Interactive Slider
# Allow range from below Tc (250K) to above Tc (350K)
interact(plot_vdW_isotherm,
        T=FloatSlider(value=304.1, min=250, max=350, step=1.0,
↪ description='Temp (K)'));

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Running locally.

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interactive(children=(FloatSlider(value=304.1, description='Temp (K)', max=
```